

POLITECNICO
MILANO 1863
DIPARTIMENTO DI ELETTRONICA
INFORMAZIONE E BIOINGEGNERIA

RF Systems

Smith Chart

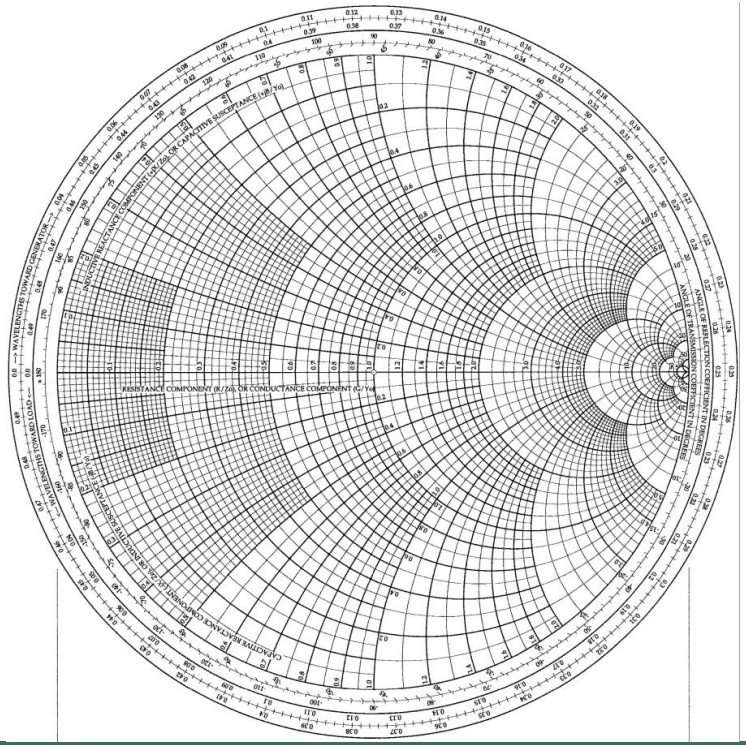
The Smith Chart

Tool to carry out computations on transmission lines

Old-fashioned: used by early RF engineers to do complex computations

But still useful today!

Easier to visualize impedances and reflection coefficients on a transmission line



Electronic versions of the Smith Chart:

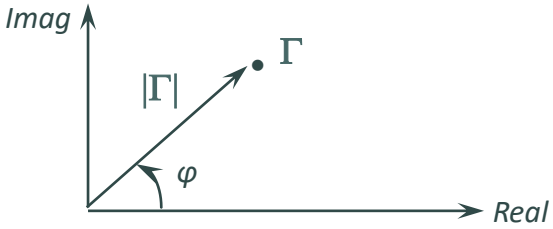
- Download from Link in Webeep (Materials→Software and Tools)

  used extensively from now on, and required for the second half of the course: Laptop needed in exams (not for midterm)

Other freeware Smith Chart tools:

- https://www.will-kelsey.com/smith_chart/
- http://www.sss-mag.com/smith.html

A graphic representation of Γ, Z, Y



Given a reference impedance Z_C, Γ is a complex number and can be represented in the complex plane: $\Gamma=|\Gamma|\, e^{j\varphi}$

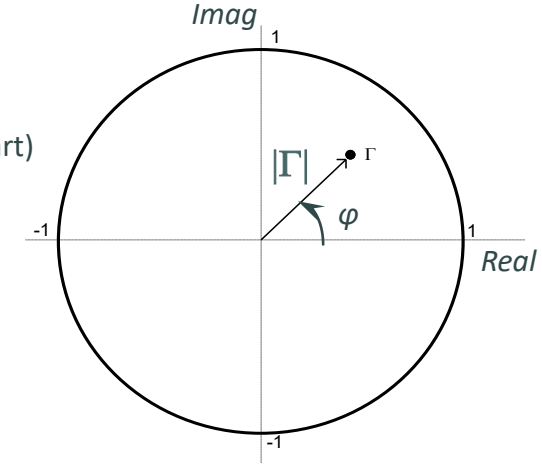
$$\Gamma = \frac{Z_n - 1}{Z_n + 1}$$

Any passive normalized impedance $z_n=Z/Z_C$ (real(z_n)>0)

corresponds to $|\Gamma|\leq 1$

Most of the time we will operate within the unit circle (boundary of the Smith Chart)

Normally Z_C is real (e.g. 50 Ω), characteristic impedance of low-loss line



To verify: $z_n=r+jx$

$|\Gamma|^2=|(r+jx-1)/(r+1+jx)|^2 = |(r+jx-1)(r+1-jx)|^2/((r+1)^2+x^2)$

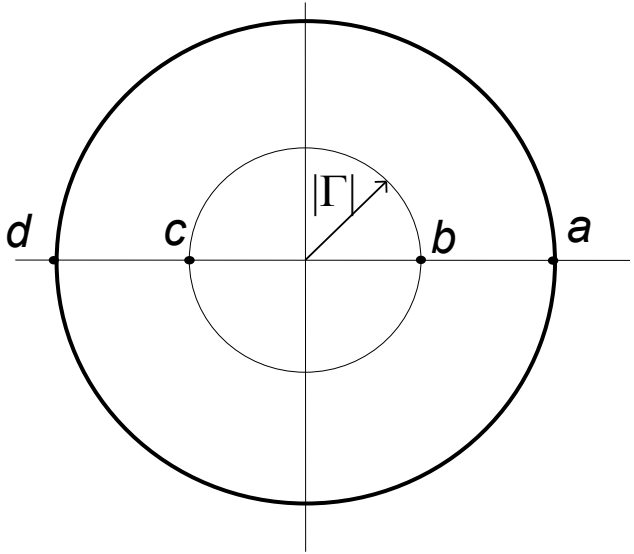
$= ((r-1)(r+1)+x^2)/((r+1)^2+x^2)=(r^2+1+x^2)/((r+1)^2+x^2)$

To have $|\Gamma|<1$: $r^2+1+x^2<(r^2+1+2r+x^2)\rightarrow r>0$

A graphic representation of Γ, Z, Y

$$\Gamma = \frac{Z_n - 1}{Z_n + 1} = \frac{1 - y_n}{1 + y_n} \qquad Z_n = \frac{Z}{Z_c} = \frac{1 + \Gamma}{1 - \Gamma} = \frac{1}{y_n}$$

$$y_n = YZ_c = 1/z_n = \frac{1 - \Gamma}{1 + \Gamma}$$



Special points:

- Γ =0 (center of the chart): $z_n=y_n=1 \rightarrow Z= Z_0$ ("matched to Z_0")
- Γ =1 (a) : $z_n=\infty$; $y_n=0 \rightarrow Z=\infty$ (open circuit)
- Γ =-1 (d) : $y_n=\infty$; $z_n=0 \rightarrow Z=0$ (short circuit)
- Maximum of voltage on a tx line (b) (Γ real and positive)
- Minimum of voltage on a tx line (c) (Γ real and negative)

$$V(z) = v^+(z) + v^-(z) = v^+(z)\big(1 + \Gamma(z)\big)$$

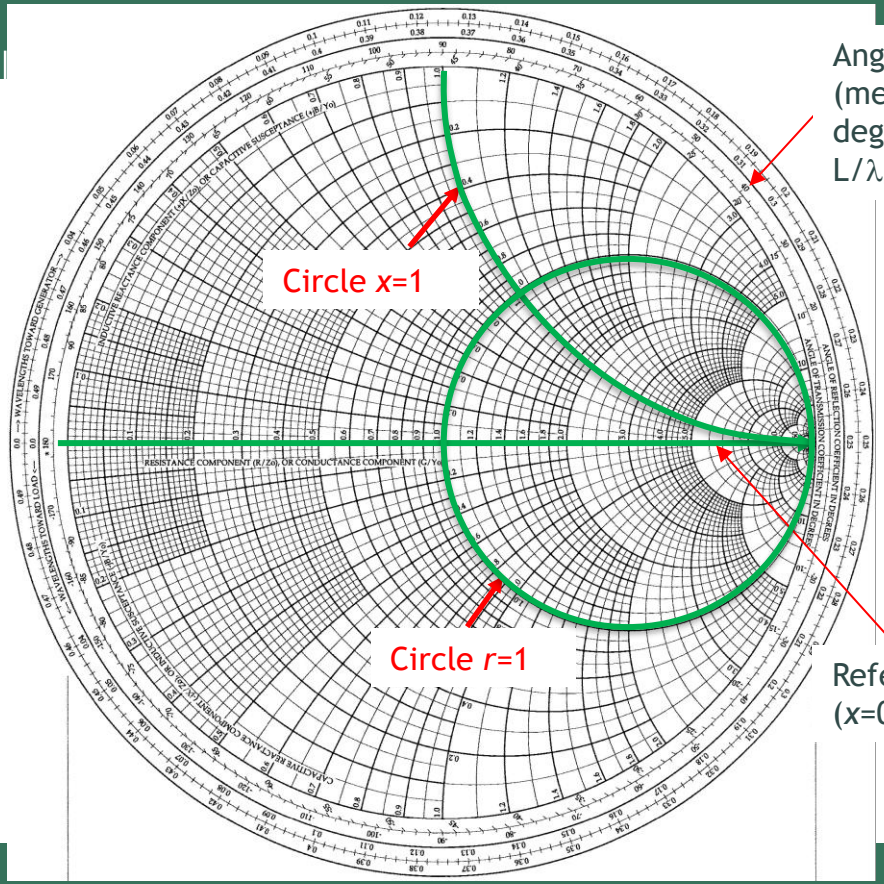
Impedance Smith Chart

- On the polar plane of Γ the curves representing the locus of points where the real (or imaginary) part of Z/Z_c is constant are defined by:

$$\mathrm{Re}\left\{\frac{Z}{Z_c}\right\} = \mathrm{Re}\left\{\frac{1+\Gamma}{1-\Gamma}\right\} = r$$

$$\mathrm{Im}\left\{\frac{Z}{Z_c}\right\} = \mathrm{Im}\left\{\frac{1+\Gamma}{1-\Gamma}\right\} = x$$

- These curves are circles, whose center and radius depend on *r* (or *x)*.
- The Smith Chart is the graphic representation of such circles.
- It allows to solve graphically several problems related to the transmission lines used in microwave circuits.

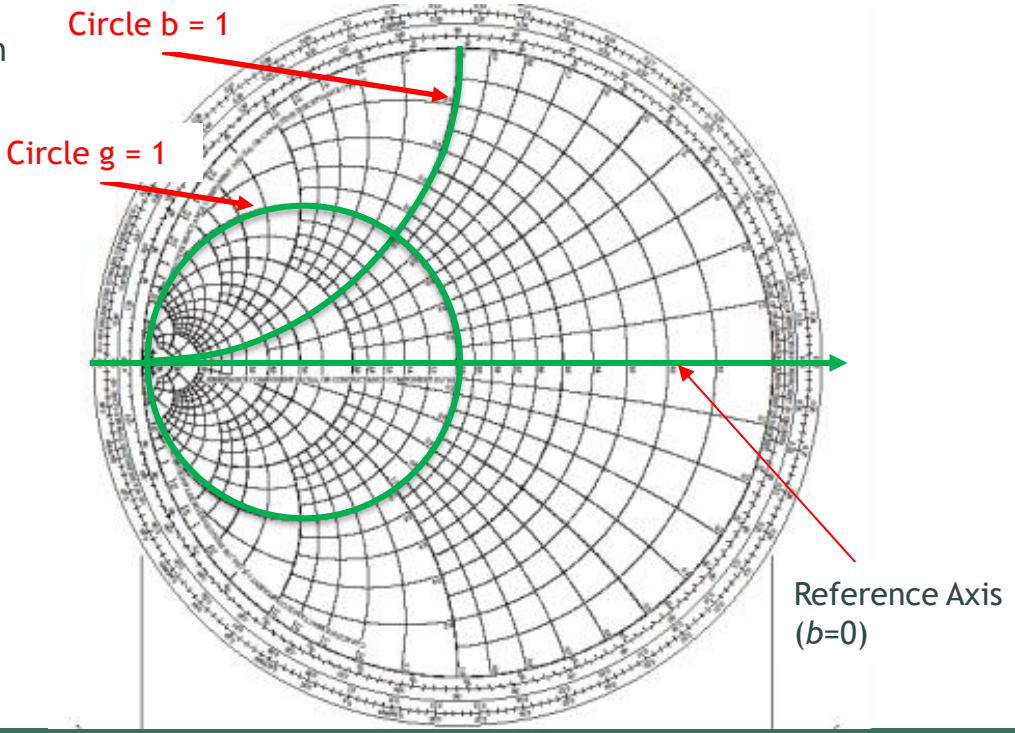


Admittance Smith Chart

Similarly, the locus of Γ corresponding to a given conductance *g* or susceptance *b* are circles:

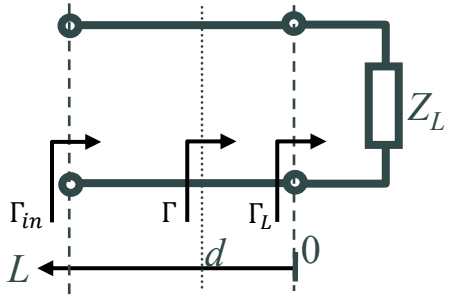
$$\mathrm{Re}\{YZ_C\} = \mathrm{Re}\left\{\frac{\Gamma - 1}{\Gamma + 1}\right\} = g$$

$$\mathrm{Im}\{YZ_C\} = \mathrm{Im}\left\{\frac{\Gamma - 1}{\Gamma + 1}\right\} = b$$



Smith Chart: Moving at |Γ|=const

Displacement: knowing Γ at one section looking in a direction, compute Γ somewhere else



On a lossless line $|\Gamma(d)|=|\Gamma_L|$

$$\Gamma(d) = \Gamma_L \cdot e^{-j2\beta d} = |\Gamma_L| e^{j\angle\Gamma_L} \cdot e^{-j2\pi \cdot \left(d/\lambda/2\right)}$$

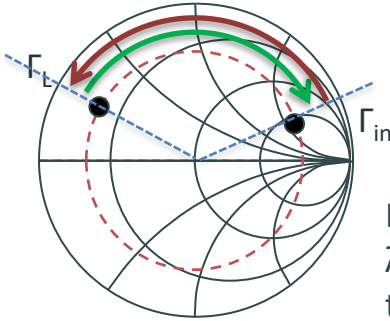
$|\Gamma(d)|=|\Gamma_L|$ → Γ(d) remains on a circle (constant radius)

- Moving against the observation direction, i.e. knowing Γ_L compute Γ_{in} (sometimes a.k.a. "moving from load toward source")

$$\angle\Gamma_{in} = \angle\Gamma_L - 2\beta L = \angle\Gamma_L - 2\phi$$ → Clockwise rotation by 2ϕ from Γ_L

- Moving along the observation direction, i.e. knowing Γ_{in} compute Γ_L (sometimes a.k.a "moving from source toward load")

$$\angle\Gamma_L = \angle\Gamma_{in} + 2\beta L = \angle\Gamma_{in} + 2\phi$$ → Counterclockwise rotation by 2ϕ from Γ_{in}

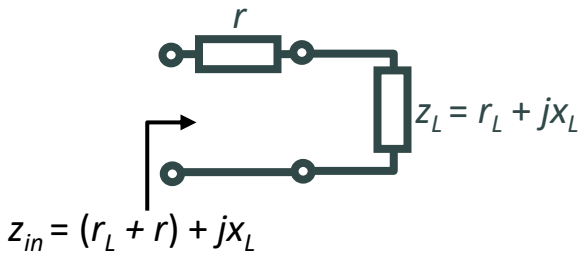


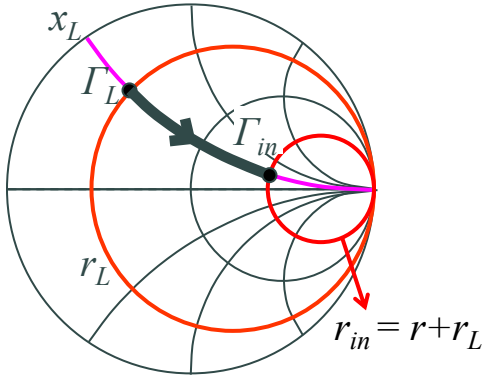
The rot. angle is twice the electrical length ϕ=βL

Displacing by λ/2 corresponds to 360° on the Smith Chart

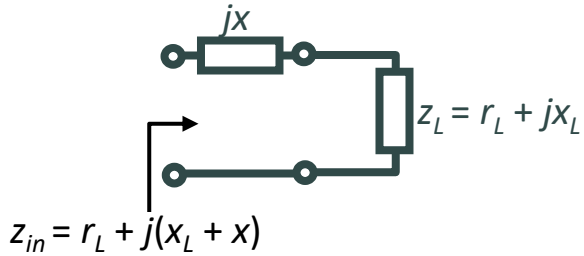
Smith Chart: Displacement at r (or x) constant

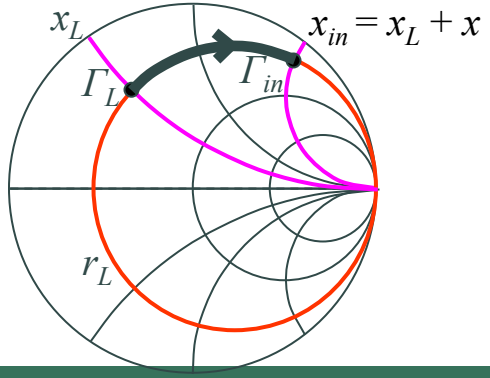
Displacement at *x* constant





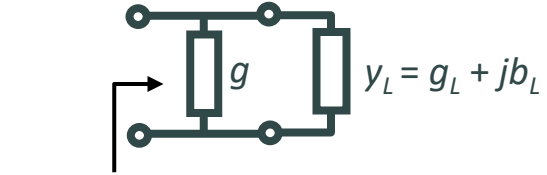
Displacement at *r* constant





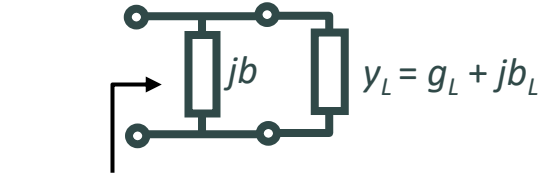
Smith Chart: Displacement at g (or b) constant

Displacement at *b*=constant

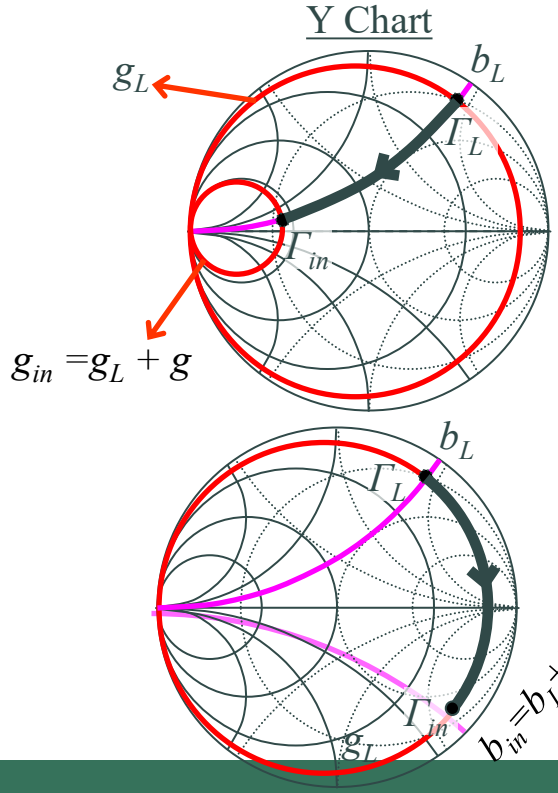


$y_{in} = (g_L + g) + jb_L$

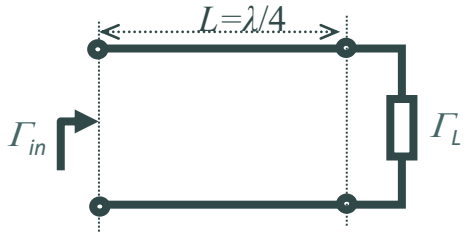
Displacement at *g*=constant



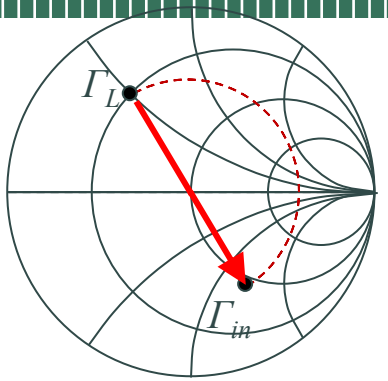
$y_{in} = g_L + j(b_L + b)$



Quarter-wave Transmission Line (ϕ=90°, L=λ/4)



$$\Gamma_{in} = \Gamma_L e^{-j4\pi\left(\frac{L}{\lambda}\right)} = \Gamma_L e^{-j\pi} = -\Gamma_L$$



$$Z_{in} = Z_c \frac{1+\Gamma_{in}}{1-\Gamma_{in}}, \qquad Z_L = Z_c \frac{1+\Gamma_L}{1-\Gamma_L}$$

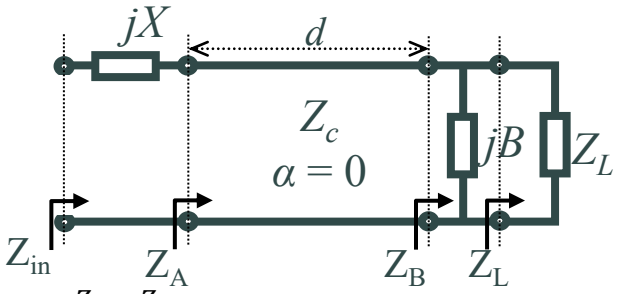
$$Z_{in} = Z_c \frac{1+\Gamma_{in}}{1-\Gamma_{in}} = Z_c \frac{1-\Gamma_L}{1+\Gamma_L} = \frac{Z_c^2}{Z_c \dfrac{1+\Gamma_L}{1-\Gamma_L}} = \frac{Z_c^2}{Z_L}$$

$$\boxed{Z_{in} = \frac{Z_c^2}{Z_L}}$$

Impedance inversion

A transmission line with length λ/4 (or λ/4+N λ/2) behaves as an IMPEDANCE INVERTER

Smith Chart: Example of graphical solution: find Z_{in}



$Z_c = 50\ \Omega$
$\varepsilon_r = 4$
$f_0 = 3$ GHz
$Z_L = (20 + j40)\ \Omega$
$(z_L = 0.4 + j0.8)$
$\beta = \omega/v = 360° \cdot f/v = 7.2$ °/mm

$B = 0.05\ \Omega^{-1}$ $(b_n = 2.5)$
$X = -80\ \Omega$ $(x_n = -1.6)$
$d = 15$ mm; $(\beta d = 108°)$

$$\Gamma_L = \frac{Z_L - Z_c}{Z_L + Z_c} = 0.62\angle 97.1°, \qquad y_L = 0.5 - j1$$

$$b_B = b_L + b_n = 1.5 \ \Rightarrow \ \Gamma_B = 0.745\angle -116.56°,$$

$$-2\beta d = -216° \ \Rightarrow \ \Gamma_A = 0.745\angle 27.43°, \qquad x_A = 2.954$$

$$x_{in} = x_A + x_n = 1.354 \ \Rightarrow \ \Gamma_{in} = 0.508\angle 31.11° \ \Rightarrow$$
$$Z_{in} = Z_c \frac{1 + \Gamma_{in}}{1 - \Gamma_{in}} \cong 95.6 + j67.7\ [\Omega]$$

